

Recipe Review Sentiment Classification Using Classical NLP and Machine Learning

An End-to-End NLP & Machine Learning Project

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PROJECT OVERVIEW

BUSINESS PROBLEM

Understanding user sentiment in recipe reviews
to improve recommendations & user experience

TECHNICAL OBJECTIVES

- Build NLP preprocessing pipeline
- Compare vectorization methods (Count vs TF-IDF)
- Train ML models (LR, NB, Linear SVC)
- Handle class imbalance with SMOTE

SKILLS DEMONSTRATED

Text preprocessing | Model selection

Imbalanced data handling | Multi-class sentiment

ABOUT THE DATASET

 SOURCE: UCI Machine Learning Repository

 18,000+ recipe reviews

FEATURES:

- Review text
- Star ratings (1-5)
- User reputation
- Engagement metrics
- Recipe info
- Timestamps

KEY INSIGHTS:

- Avg review length: 204 characters
- Reviews generally short and informal

DATA PREPARATION & EXPLORATION

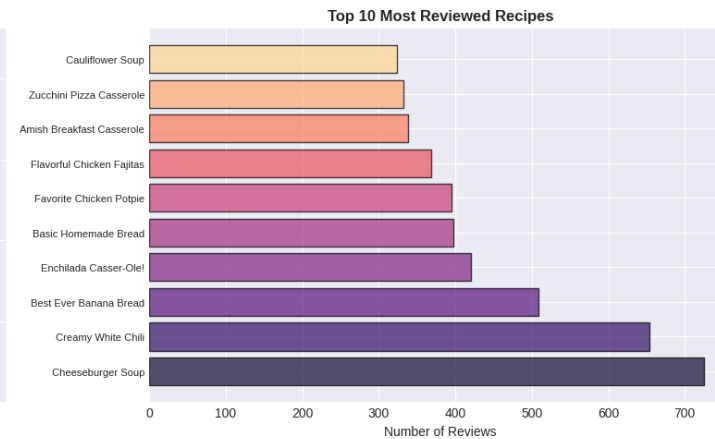
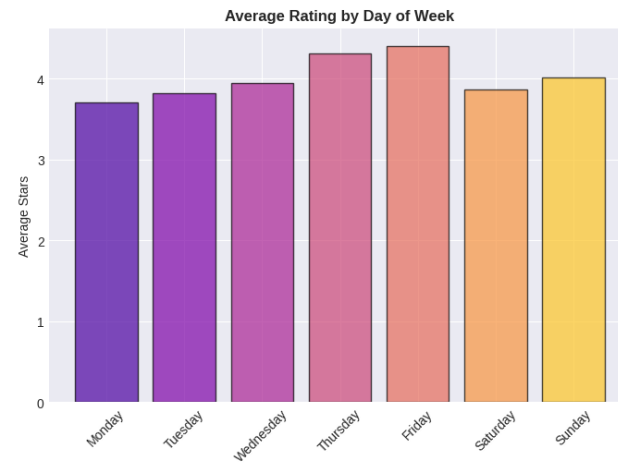
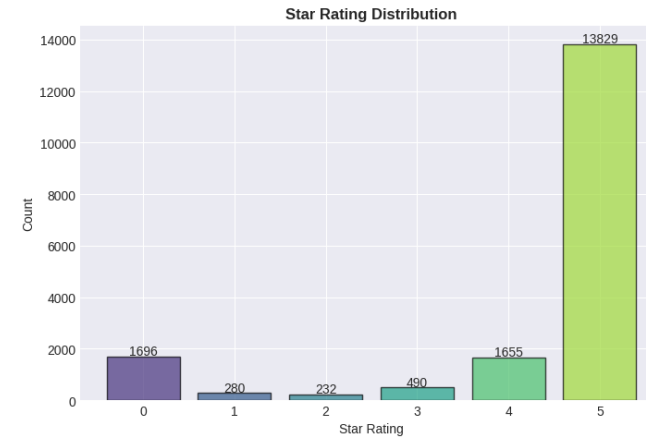
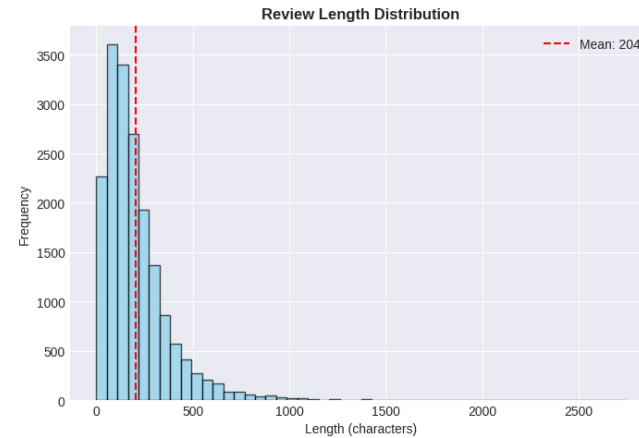
DATA QUALITY ASSESSMENT

- ✅ Missing values: Only 2 (0.01%) - filled with “ “
- ✅ Duplicates: None found
- ✅ Removed: 'Unnamed: 0' index column

FEATURE ENGINEERING

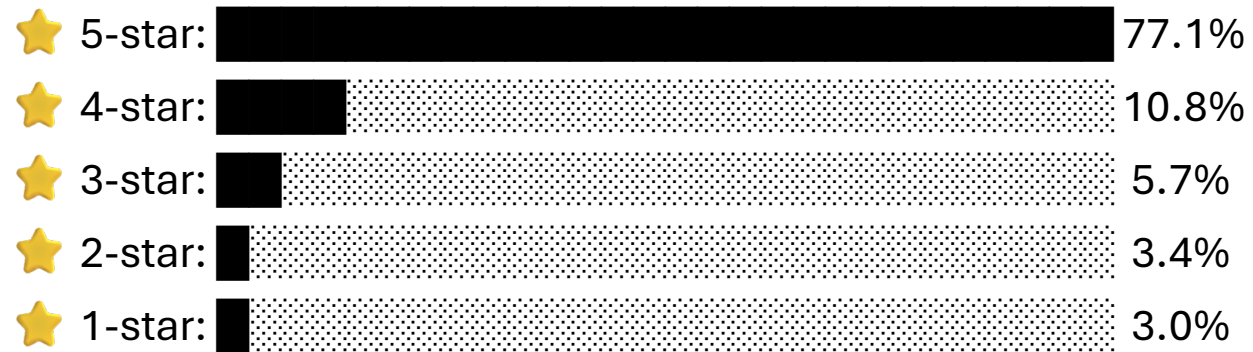
- 📅 Temporal: year, month, day_of_week
- 📝 Review length: avg 204 chars, median 161 chars

DATA PREPARATION & EXPLORATION



CLASS IMBALANCE ANALYSIS

CLASS IMBALANCE ⚠️



IMPACT:

- ✗ Models biased toward positive reviews
- ✗ Minority classes poorly predicted
- ✓ SMOTE needed for fair classification

NLP PREPROCESSING PIPELINE

TEXT PROCESSING WORKFLOW

Raw Text → Tokenization → Lowercasing → Filter → Stopword Removal → Lemmatization → Processed Tokens

EXAMPLE

ORIGINAL:

"I tweaked it a little, removed onions because of onion haters in my house"

↓ PROCESSED ↓

['tweak', 'little', 'remove', 'onions', 'onion', 'haters', 'house', 'use', 'italian', 'season']

VECTORIZATION METHODS

 CountVectorizer: Raw frequency counts

 TF-IDF: Term frequency-inverse document frequency

 Configuration: min_df=2, max_features=2000

VADER SENTIMENT ANALYSIS

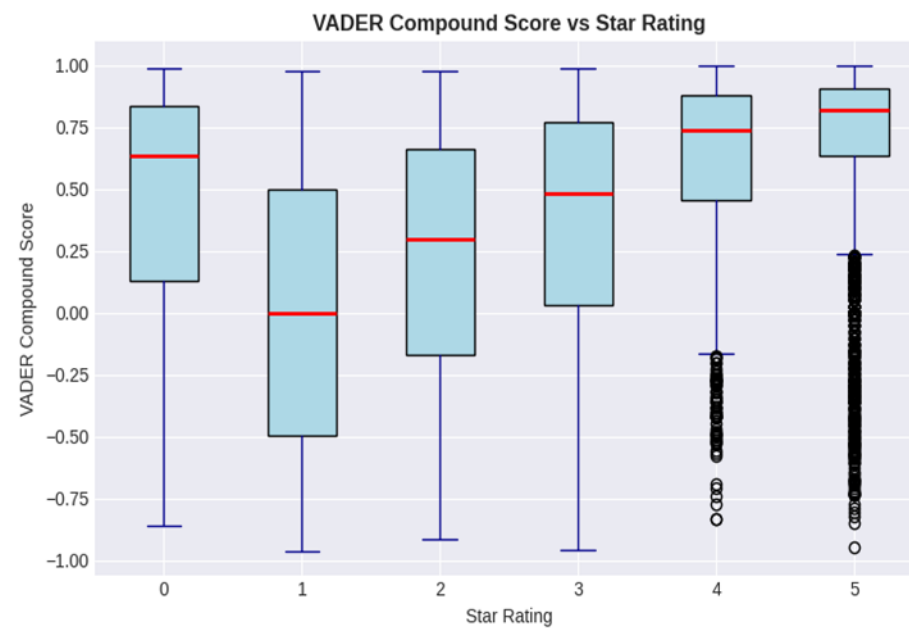
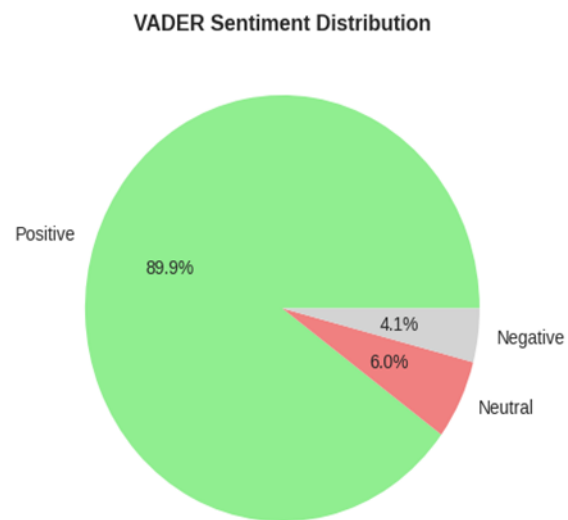
VADER Sentiment Distribution:



VADER Score vs Star Rating:

- Strong alignment with user ratings
- Higher scores for 4-5 star reviews
- Validates sentiment approach

VADER SENTIMENT ANALYSIS



6-CLASS CLASSIFICATION RESULTS

TARGET: 1-5 Star Ratings (6 classes)

 PERFORMANCE:

Model	Accuracy	Macro F1	Wtd F1
Logistic + Count	75.4%	0.325	0.718
Logistic + TF-IDF	77.5%	0.278	0.709
Naive Bayes + Count	72.1%	0.325	0.707

KEY INSIGHT

- Accuracy looks reasonable (~75%)
- BUT Macro F1 is very low (0.278-0.325)
- Minority classes (1-4 stars) are poorly predicted
- Class imbalance severely impacts performance

SMOTE BALANCING

3-CLASS WITH SMOTE

LABEL MAPPING:

- 1-2 stars → Negative
- 3 stars → Neutral
- 4-5 stars → Positive

SMOTE IMPACT:

Class	Before	After
Positive	14,145	14,145
Neutral	912	14,145 
Negative	1,035	14,145 

IMPROVEMENTS:

- 40-60% F1 increase for minority classes
- Fairer model performance

FINAL RESULTS

FINAL MODEL COMPARISON

 PERFORMANCE (with SMOTE):

Model	Accuracy	Wtd F1
Logistic Regression	75.8%	0.758
Naive Bayes	71.3%	0.714
Linear SVC 🏆	75.4%	0.777

 CLASS-WISE F1 SCORES:

Class	F1 Score
Negative	0.70
Neutral	0.37 ⚠️
Positive	0.77

 BEST MODEL: Linear SVC + SMOTE

KEY INSIGHTS & IMPACT

TOP FINDINGS:

1. CLASS IMBALANCE IS CRITICAL
 - SMOTE improved minority F1 by 40-60%
2. MODEL SELECTION MATTERS
 - Linear SVC achieved best performance
 - 75.4% accuracy, 0.777 weighted F1
3. NEUTRAL CLASS IS CHALLENGING
 - Lowest F1 (0.37) due to ambiguous reviews

BUSINESS APPLICATIONS:

- Identify dissatisfied users for service
- Promote highly-rated recipes
- Detect negative sentiment trends
- Improve recommendation systems

CHALLENGES & FUTURE WORK

CHALLENGES:

- Severe class imbalance (77% 5-star)
- Short, informal reviews (avg 204 chars)
- Neutral class difficult to classify

FUTURE WORK:

- Transformer models (BERT, DistilBERT)
- Alternative balancing techniques
- Recommendation systems using embeddings
- Aspect-based sentiment analysis

CONCLUSION

✅ ACHIEVEMENTS:

- Built complete end-to-end NLP pipeline
- Processed 18,000+ recipe reviews
- Compared multiple ML models
- Addressed class imbalance with SMOTE
- Achieved 75%+ accuracy & 0.777 weighted F1

 BEST MODEL: Linear SVC + SMOTE

💡 KEY TAKEAWAY:

Classical ML remains effective when handling class imbalance properly. SMOTE significantly improves minority class performance.

🔮 NEXT STEPS:

Explore deep learning & transformer models for further improvement.

REFERENCES

 DATASET:

[UCI Machine Learning Repository - Recipe Reviews](#)

 COURSES:

- [NLP with Classification \(Coursera\)](#)
- [ML for Beginners \(Microsoft\)](#)

 DOCUMENTATION:

- [NLTK](#) • [Scikit-Learn](#)

 BOOKS:

[NLP with Python \(Bird, Klein & Loper\)](#)

 MENTORSHIP:

Dr. Raji Ghawi - ReDI Connect

THANK YOU